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Developing design formulae for flexural buckling of stainless steel columns using machine learning techniques

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2.1 Structural stability and the mechanics of non-linear materials

2.1.1 Column buckling and the normalised slenderness approach

Perfect columns are idealised structural members that are perfectly straight and exhibit elastic–perfectly plastic behaviour. The strength of such a member is bounded by two limiting modes of failure: geometric instability and material yielding. The former is governed by the Euler critical buckling load, which for a pin-ended column is

$$N_{\mathrm{cr}} = \frac{\pi^2 EI}{L^2}, \tag{2.1}$$

where E is Young's modulus, I is the second moment of area and L is the member length (Timoshenko and Gere, 2009). The latter is governed by the squash load $N_{\mathrm{y}} = Af_{\mathrm{y}}$, where A is the cross-sectional area and f_{y} is the yield stress. Together these define the normalised slenderness $\bar{\lambda}$, which collapses the column's geometry, material and length into a single dimensionless parameter, while the ultimate load N_{u} is normalised by the squash load to give the buckling reduction factor χ:

$$\bar{\lambda} = \sqrt{\frac{N_{\mathrm{y}}}{N_{\mathrm{cr}}}}, \qquad \chi = \frac{N_{\mathrm{u}}}{N_{\mathrm{y}}}. \tag{2.2}$$

As illustrated in Figure 2.1, the theoretical maximum strength of a perfect column forms a sharp χ–$\bar{\lambda}$ baseline: a pure squashing plateau ($\chi = 1$) for stocky columns and an elastic Euler hyperbola ($\chi = \bar{\lambda}^{-2}$) for slender columns, the two regimes meeting at $\bar{\lambda} = 1$ (Trahair, 2008).

Real columns, however, never reach the perfect baseline because they carry imperfections such as initial crookedness, residual stresses, and load eccentricities. As a result, failure is observed below the theoretical bounds described, particularly at intermediate slenderness ($\bar{\lambda} \approx 0.7$–$1.5$) where instability and yielding interact.

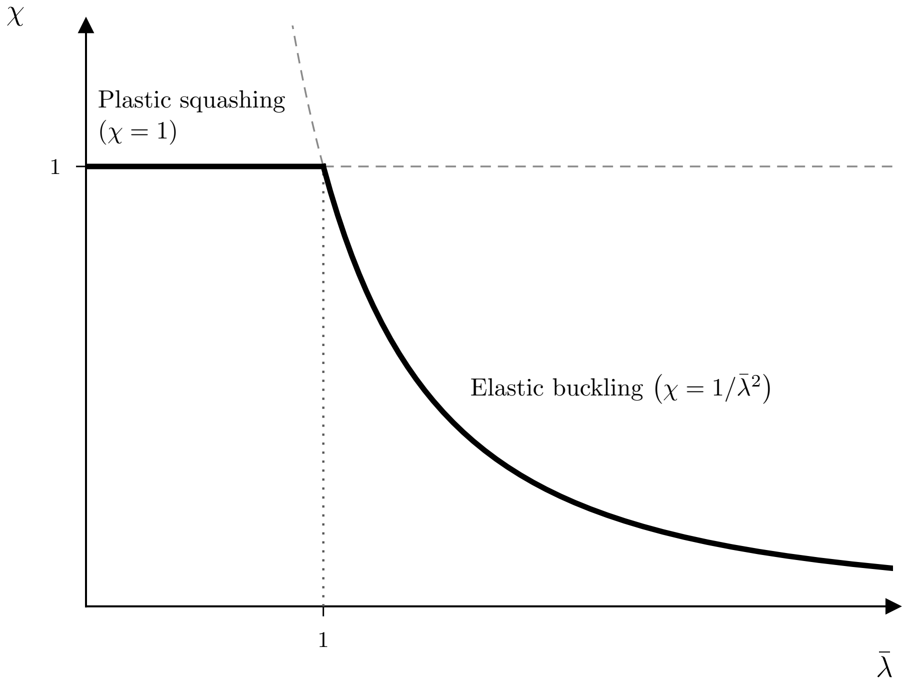

Figure 2.1: Buckling reduction factor χ against normalised slenderness $\bar{\lambda}$ for a perfect, elastic–perfectly plastic column: the lower envelope of the squash plateau ($\chi = 1$) and the Euler hyperbola ($\chi = \bar{\lambda}^{-2}$), which meet at $\bar{\lambda} = 1$.

2.2 Design philosophies for non-linear material buckling

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2.5 Machine learning for strength prediction in structural members

and Zhang (2021) compared seven algorithms for predicting the capacity of cold-formed stainless steel tubular columns and found tree-ensemble methods to be the most accurate: the random forest achieved a mean predicted-to-test capacity ratio of 1.01 (COV $\approx$ 0.08), against a mean of 0.89 for the Eurocode method. Similar findings have been reported across member types. For circular hollow section stub columns, the EC3 resistance was found to be substantially conservative, underpredicting test capacities by roughly 30%, whereas the best machine-learning model reproduced them almost exactly (mean ratio 1.00) (Abarkan et al., 2024). Equally, for concrete-filled stainless steel tubes, Tran, Ahmed, and Gohery (2023) found a gradient-boosting model which reduced a similar underprediction to a mean ratio of about 1.04. These findings echo the limitation established in Section 2.2.1: flexible, data-driven models capture the non-linear material and geometric response more faithfully than expressions built on fixed, pre-calibrated constants.

This study aims to draw on these developments for a more specific purpose than direct strength prediction. The aim is to extend the correction factor of Section 2.3.2 from a univariate function of $\bar{\lambda}$ into one that reflects the physical drivers identified in section 2.4. Instead of assuming a fixed functional form, the model learns the relationship between the chosen descriptors and the observed strength directly from the experimental data. The interpretability tools introduced later in this section then show which descriptors genuinely carry predictive value. In this sense, machine learning is used to inform an extended, mechanically grounded correction factor, not to replace the formulation in Section 2.3 with a black-box predictor.

Given the scarcity of experimental stability data for stainless steel and its specific tabular, heterogeneous format, the stated goal must be pursued within a strict data regime. The model choice and the manner in which data are partitioned for validation are two important considerations, which are explored in the following subsections.

2.5.2 Evaluation of model architecture for small tabular datasets

Experimental data on member stability, particularly for stainless steel, have a particular structure that dictates model choice. Every datum is the product of a physical test taken to failure; consequently, independent studies typically comprise only a handful of experimental tests, which are later used to augment and validate a much larger dataset via finite-element (FE) simulations (e.g., (Buchanan, Real, and Gardner, 2018; Yuan et al., 2015; Sun et al., 2020)). The resulting data are tabular and heterogeneous, mixing continuous quantities such as material parameters and geometry with categorical descriptors such as fabrication route and cross-section type. A broad range of algorithms have been applied to problems of this kind, but three recur as the dominant choices for structural capacity prediction: artificial neural networks, support vector regression, and tree ensembles.

An artificial neural network (ANN), in its common multi-layer perceptron form, maps

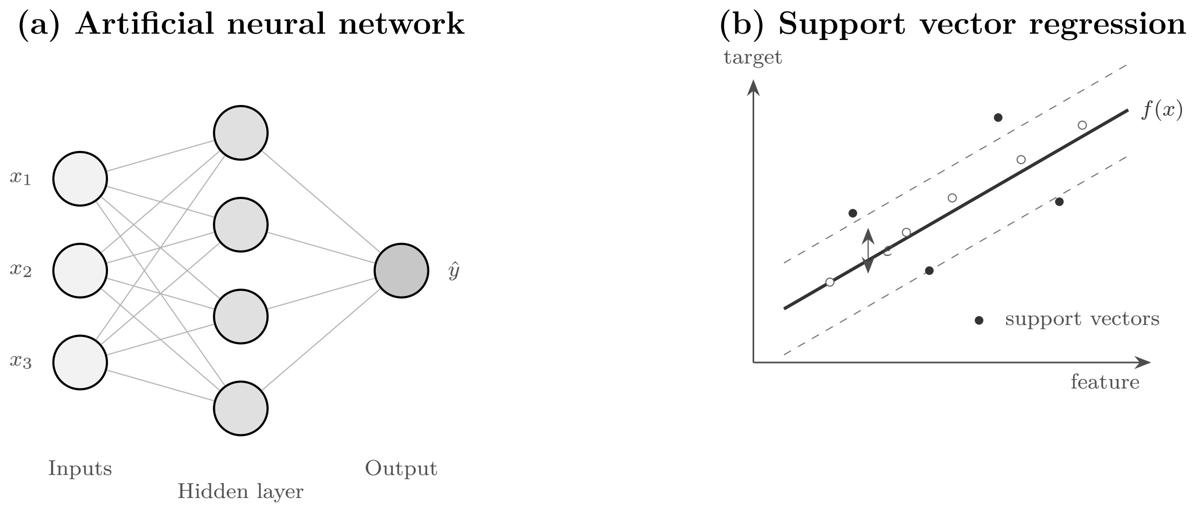

Figure 2.2: The two flexible, high-capacity architectures most widely applied to structural capacity prediction. (a) A multi-layer perceptron maps inputs through hidden layers of weighted sums and non-linear activations. (b) Support vector regression fits a function within an ε-insensitive tube, determined by a subset of support vectors.

inputs to outputs through successive layers of interconnected nodes, each applying a weighted sum followed by a non-linear activation, with the weights fitted by gradient descent (Goodfellow, Bengio, and Courville, 2016). Support vector regression (SVR) instead maps the inputs into a higher-dimensional space through a kernel function and fits a regression surface within a fixed error tolerance, its complexity governed by regularisation (Smola and Schölkopf, 2004). These two architectures are contrasted schematically in Figure 2.2. The primary advantage of both, and the reason they have been applied so widely and successfully to structural capacity prediction, lies in their ability to map complex, high-dimensional interactions without requiring a predefined mathematical form (Thai, 2022). This flexibility is particularly valuable when predicting flexural buckling, as it allows the algorithms to organically capture the deeply coupled effects of geometric imperfections and continuous strain-hardening material behaviour.

However, their suitability weakens significantly outside the large-data regime. The high predictive accuracy of deep learning models in structural engineering is overwhelmingly reliant on massive synthetic datasets, typically generated through extensive FE studies rather than physical testing (Thai, 2022). When constrained to small, purely experimental datasets, these models become unreliable. ANNs, which carry a vast number of free parameters, tend to overfit sparse data and consequently overlook the underlying physical mechanics (Goodfellow, Bengio, and Courville, 2016; Brigato and Iocchi, 2020). Conversely, the generalisation capability of SVR is particularly sensitive to the choice of kernel and its hyperparameter tuning (Hastie, Tibshirani, and Friedman, 2009).

This instability has been observed on experimental stainless steel data. In a study by Xu, Zheng, and Zhang (2021), which aimed to establish a unified capacity prediction

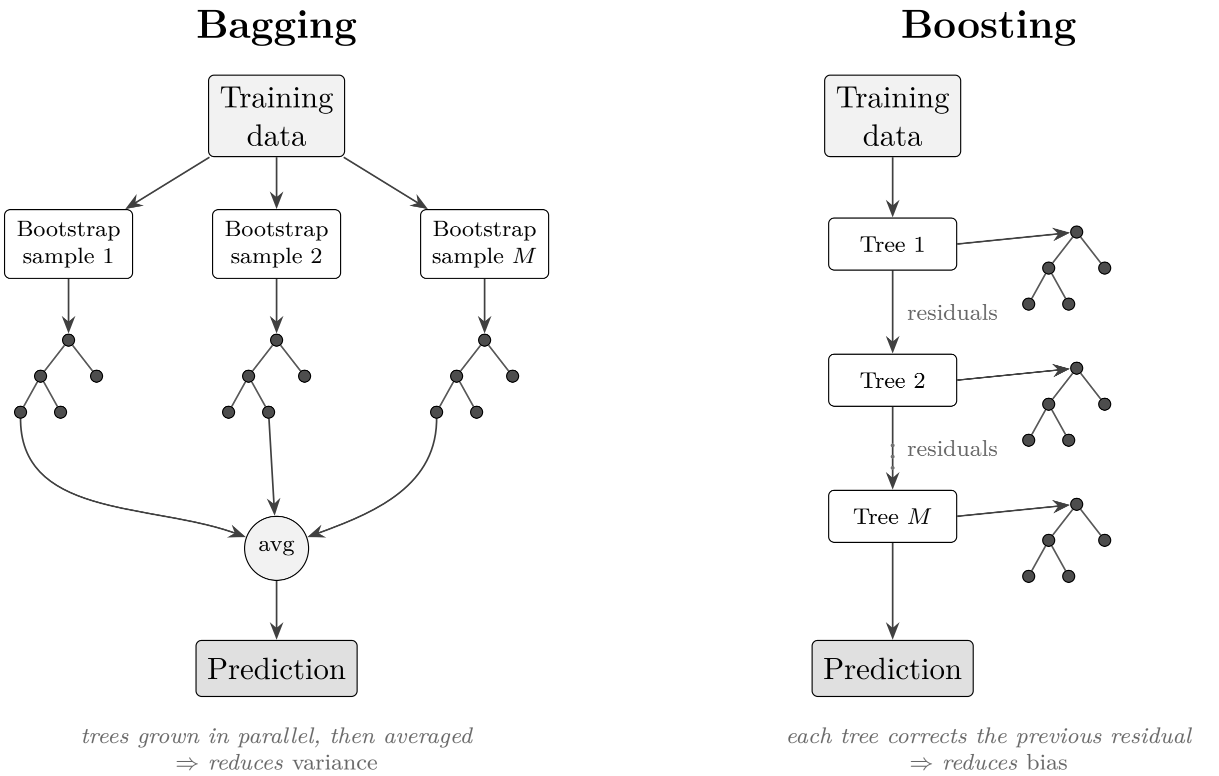

Figure 2.3: The two tree-ensemble families. Bagging grows trees in parallel on independent bootstrap resamples and averages them, reducing variance; boosting grows trees sequentially, each fitted to the residuals of the current ensemble, reducing bias.

method for 322 cold-formed stainless steel tubular columns, both the ANN and the SVR were reported as highly sensitive to parameter tuning and unstable in accuracy. The SVR, in particular, deteriorated sharply from training to testing, with its predicted-to-test scatter rising roughly fivefold, while the neural network was deliberately kept small precisely because the sample was too limited to support a larger one.

Tree-ensemble methods inherently bypass the limitations of deep networks and kernel methods on small tabular datasets. By partitioning the feature space with simple, axis-aligned splits, decision trees can capture non-linear response surfaces without extensive hyperparameter tuning or feature scaling (Grinsztajn, Oyallon, and Varoquaux, 2022). These ensembles fall into two primary families, illustrated schematically in Figure 2.3. Bagging methods, such as random forests, grow many trees independently on bootstrap resamples of the training data and average their predictions, reducing *variance* through decorrelation (Breiman, 2001). Boosting methods, such as gradient boosting, grow trees sequentially, each fitted to the residual error of the current ensemble, reducing *bias* through successive correction (Friedman, 2001). Modern gradient-boosting implementations further introduce internal regularisation, highly efficient split-finding algorithms, and native categorical handling (Chen and Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2019).

Empirical evidence strongly favours tree ensembles for structural capacity prediction on limited datasets. In the previously mentioned study by Xu, Zheng, and Zhang (2021), random forest and XGBoost significantly outperformed ANNs and SVRs, rectifying conservative Eurocode predictions, which systematically underestimated capacity with a mean predicted-to-test ratio of 0.89. A subsequent study of 280 circular hollow section (CHS) columns corroborated these findings, with the random forest achieving a highly accurate mean ratio of 1.00 and a standard deviation of 0.083 (Xu, Zhang, and Zheng, 2021). This pattern is consistent across adjacent structural problems: gradient boosting outperformed ANNs and tuned SVRs for concrete-filled tubes (Megahed, Mahmoud, and Abd-Rabou, 2023), and CatBoost yielded the highest accuracy on a sparse dataset of 252 gusset-plate connections by exploiting its native categorical treatment (Rahman et al., 2024). These domain-specific results align precisely with the broader machine learning consensus: tree ensembles remain the state-of-the-art architectures for small-to-medium tabular data, whereas neural networks demand substantially larger samples to achieve parity (**shwartzziv2021**; Grinsztajn, Oyallon, and Varoquaux, 2022).

On this basis, the tree-ensemble family is adopted for Stage 1 of this study. The three leading gradient-boosting implementations—XGBoost, LightGBM, and CatBoost—are deployed as the primary architectures due to their robust internal regularisation and bias-reducing mechanics. CatBoost is particularly valuable here, as its native categorical handling prevents dimensional inflation when incorporating non-continuous features such as fabrication route and section type. The random forest is retained as a robust benchmark given its proven efficacy in comparable stainless steel studies, and all candidates are compared under the grouped cross-validation scheme detailed in section 2.5.3. Crucially, the ultimate aim is not to supplant the established mechanics-based formulations of section 2.3, but rather to harness the ensemble's predictive accuracy to distil a physically meaningful correction factor.

2.5.3 Cross-validation for small, grouped datasets

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